

DOCTORAL RESEARCHER · COMPUTER SCIENCE

Mert Onur Cakiroglu.

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Abstract—I teach machines to read the structure hidden in sequences. My research turns video, electricity grids, glucose traces, and proteins into symbolic graphs (de Bruijn graphs, borrowed from genomics) so that neural models can reason over them. Eleven works spanning *Scientific Reports*, *IEEE Access*, *Machine Learning: Science and Technology*, *IEEE OAJPE*, and an ICML 2025 workshop; first author on ten.

Index Terms—de Bruijn graphs · video understanding · time-series forecasting · self-supervised & federated learning · graph neural networks · representation learning · reinforcement learning · PyTorch

EDUCATION

- 2023– **Ph.D., Computer Science** · Indiana University Bloomington, Luddy School of Informatics, Computing, and Engineering
Advisor: Prof. Mehmet M. Dalkilic · Co-advisor: Dr. Hasan Kurban
- 2017–21 **B.S., Computer Science** · TOBB University of Economics and Technology, Ankara

PUBLICATIONS

Peer-Reviewed Journals

- [1] **M. O. Cakiroglu**, I. B. Altun, S. R. Fahim, H. Kurban, M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin. **An Extended Frequency-Improved Legendre Memory Model for Enhanced Long-Term Electricity Load Forecasting.** *IEEE Open Access Journal of Power and Energy*, 12, 691–701, 2025.
 DOI 10.1109/OAJPE.2025.3615513
- [2] **M. O. Cakiroglu**, H. Kurban, E. Buxton, M. Dalkilic. **A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting Using TimesNet.** *IEEE Access*, 13, 123182–123198, 2025.
 DOI 10.1109/ACCESS.2025.3588507 · CODE github.com/KurbanIntelligenceLab/dBGTime-Series-Library
- [3] **M. O. Cakiroglu**, H. Kurban, L. Aljihmani, K. Qaraqe, G. Petrovski, M. M. Dalkilic. **A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients Using an Extended de Bruijn Graph.** *Scientific Reports*, 14, 31251, 2024.
 [NATURE PORTFOLIO] DOI 10.1038/s41598-024-82649-4 · CODE github.com/KurbanIntelligenceLab/dBG-Hypoglycemia-Forecast
- [4] **M. O. Cakiroglu**, H. Kurban, P. Sharma, M. O. Kulekci, E. K. Buxton, M. Raeeszadeh-Sarmazdeh, M. M. Dalkilic. **An Extended de Bruijn Graph for Feature Engineering Over Biological Sequential Data.** *Machine Learning: Science and Technology*, 5(3), 035020, 2024.
 [IF 6.8 · OPEN ACCESS] DOI 10.1088/2632-2153/ad5fde

Conferences & Workshops

- [5] **M. O. Cakiroglu**, I. B. Altun, M. Dalkilic, E. Buxton, H. Kurban. **Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting.** *ICML 2025, 1st Workshop on Foundation Models for Structured Data*, Vancouver, 2025.
 [ICML WORKSHOP] ARXIV 2505.22768 · CODE github.com/KurbanIntelligenceLab/MultdBG-Time-Series-Library
- [6] **M. O. Cakiroglu**, I. B. Altun, S. R. Fahim, H. Kurban, M. M. Dalkilic, R. Atat. **De Bruijn Graph-Enhanced Time Series Models for Electricity Load Forecasting.** *International Symposium on Signals, Circuits and Systems (ISSCS)*, Iasi, Romania, pp. 1–4, 2025.
 DOI 10.1109/ISSCS66034.2025.11105646

Conferences & Workshops (continued)

[7] M. O. Cakiroglu, H. Kurban, E. K. Buxton, M. Dalkilic. **A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting Using TimesNet (Extended Abstract).** *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, San Diego, pp. 1–3, 2024.
DOI 10.1109/DSAA61799.2024.10722826

Preprints

[8] M. O. Cakiroglu, M. Dalkilic, H. Kurban. **The Spectrum Is Not Enough: When Context Helps Time-Series Forecasting.** *arXiv:2607.13006*, 2026.
CODE github.com/KurbanIntelligenceLab/SINE

[9] M. O. Cakiroglu, Z. Lu, M. M. Dalkilic, H. Kurban. **Auditing Generalization in AI-Generated Video Detection: A Six-Control Protocol and the VidAudit Toolkit.** *arXiv:2606.31004*, 2026.
CODE github.com/KurbanIntelligenceLab/vidaudit

[10] I. B. Altun, M. O. Cakiroglu, E. Buxton, M. Dalkilic, H. Kurban. **LGQ: Learnable Geometric Quantization for Image Tokenization.** *arXiv:2602.16086*, 2026.
CODE github.com/KurbanIntelligenceLab/LGQ

[11] M. O. Cakiroglu, I. B. Altun, Z. Lu, M. Dalkilic, H. Kurban. **Temporal Realism Evaluation of Generated Videos Using Compressed-Domain Motion Vectors.** *arXiv:2511.13897*, 2025.
[IN SUBMISSION] CODE github.com/KurbanIntelligenceLab/Motion-Vector-Learning

RESEARCH APPOINTMENTS

2024	Research Associate · Texas A&M University at Qatar, Doha <i>Self-supervised learning frameworks for video and federated video learning in the compressed domain, preserving privacy and reducing communication overhead.</i>
2023–	Student Researcher · Kurban Intelligence Lab <i>De Bruijn graph representations for time series, clinical forecasting, protein classification, and video understanding.</i>

TEACHING

2023–	CSCI-C 200 · Introduction to Computers and Programming · Associate Instructor, Indiana University <i>with Prof. Mehmet M. Dalkilic</i>
2024	CSCI-B 657 · Computer Vision · Associate Instructor, Indiana University <i>with Prof. David Crandall</i>

RECOGNITION

2023	Luddy Doctoral Associate Instructor Fellowship · Luddy School of Informatics, Computing, and Engineering
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OPEN SOURCE

2024–	Maintainer of the research code behind refs [1]–[11] at github.com/KurbanIntelligenceLab
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INDUSTRY

2021–23	Full-Stack Software Developer · Innova IT Solutions <i>Built MARS, a centralized fault-management system for national telecommunication networks: Spring Boot microservices over PL/SQL and MongoDB, React and Vaadin front ends, delivered in an agile environment.</i>
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INTERESTS · ELECTRIC GUITAR · LINUX & OPEN-SOURCE SOFTWARE